

# Revision — sections of polyhedra and orthogonal projection

Three rules build any section; one shadow measures any angle.

LESSON 65–66

2 × 40 MIN

GEOMETRY 10, CHAPTER IV REVIEW

P1 SPATIAL REVIEW

LESSON CLOCK

15'

30'

20'

20'

15'

|                     |        |
|---------------------|--------|
| Three rules         | 15 min |
| Building a section  | 30 min |
| Naming the shape    | 20 min |
| Areas by projection | 20 min |
| Homework            | 15 min |

LEARNING OBJECTIVES

- Construct the section of a cube or prism by a plane through three points.
- Use the trace of a cutting plane on a face of the solid.
- Compute the area of a section by projecting it onto the base.
- Use orthogonal projection to find lengths and angles.

TEXTBOOK

|           |                               |
|-----------|-------------------------------|
| National  | pp. 267–276                   |
| Cambridge | Pure Mathematics 1, pp. 81–84 |
| Workbook  | Revision set G2               |

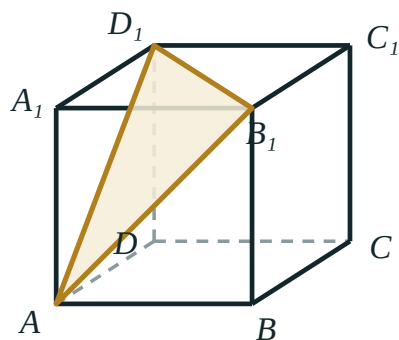
01

## Explanation

### Three rules

EVERYTHING A SECTION NEEDS

- Two points of the cutting plane on the **same face** join up — the segment between them is part of the section.
- The cutting plane meets two **parallel faces** in **parallel** lines.
- Where the cutting plane meets the plane of a face outside the solid, that point is on the **trace**, and can be used to reach a further face.



the section  
A B<sub>1</sub> D<sub>1</sub> is  
an equilateral  
triangle

*A plane through three points of a cube — each  
segment lies in one face.*

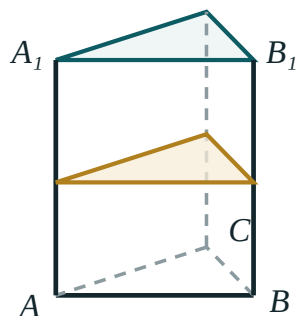
A SECTION IS A POLYGON, AND ITS SIDES LIE IN FACES

Every side of a section is the intersection of the cutting plane with **one face**. A drawn line that crosses from one face to another is not a side of anything — it is a mistake.

## Building a section

The method, for a plane through three given points of a cube:

1. join any two points that lie on the same face — rule 1;
2. extend that line to meet the plane of an adjacent face outside the solid — rule 3;
3. join the new point to the third given point, and take the part inside the solid;
4. use rule 2 on parallel faces to finish, and check every side lies in exactly one face.



a section parallel  
to the base is  
congruent to it

every cross-section of a prism repeats the base

*The same three rules on a prism — a plane cut across  
the lateral faces.*

| CUT OF A CUBE                | SHAPE                | NOTE                       |
|------------------------------|----------------------|----------------------------|
| parallel to a face           | square               | congruent to the face      |
| through two opposite edges   | rectangle            | a diagonal section         |
| through three face-diagonals | equilateral triangle | side $a\sqrt{2}$           |
| through six edge-midpoints   | regular hexagon      | side $\frac{a\sqrt{2}}{2}$ |

Naming the shape

The shape of the section is decided by how many faces the plane crosses — one side per face crossed.

| FACES CROSSED | SECTION       | EXAMPLE IN A CUBE           |
|---------------|---------------|-----------------------------|
| 3             | triangle      | a corner cut off            |
| 4             | quadrilateral | a plane parallel to an edge |
| 5             | pentagon      | an oblique cut              |
| 6             | hexagon       | through six edge-midpoints  |

A cube has six faces, so its sections have at most six sides. A tetrahedron has four, so its sections are triangles or quadrilaterals — never more.

PARALLEL FACES FORCE PARALLEL SIDES

In a cube the faces come in three parallel pairs, so a hexagonal section has its opposite sides parallel — and the regular one, through the six edge-midpoints, has all six sides equal as well.

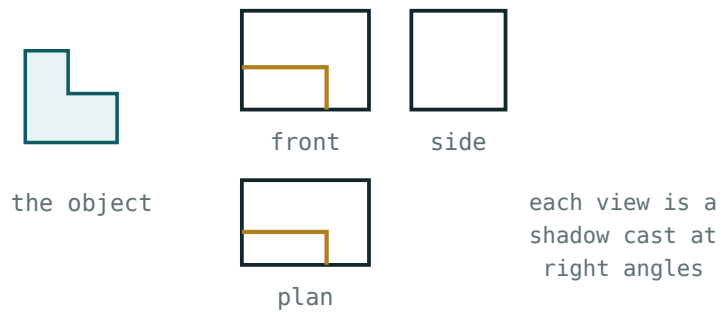
Areas by projection

When a section is awkward to measure directly, project it onto the base.

$$S_{\text{projection}} = S_{\text{section}} \cdot \cos \theta$$

where  $\theta$  is the dihedral angle between the two planes. Read backwards, this finds the section from a shape that is easy to measure:

$$S_{\text{section}} = \frac{S_{\text{projection}}}{\cos \theta}$$



three orthogonal projections determine the solid

*The shadow is shorter than the object by exactly the factor  $\cos \theta$ .*

**Example.** A plane cuts a cube of edge 6 through  $A$ ,  $B$  and  $C_1$ . Its projection on the base is the triangle  $ABC$ , of area 18; the dihedral angle is  $45^\circ$ ; so the section has area

$$\frac{18}{\cos 45^\circ} = 18\sqrt{2} \approx 25.5.$$

#### THE PROJECTION IS ALWAYS THE SMALLER

Since  $\cos \theta \leq 1$ , a shadow is never larger than what casts it. If your section comes out *smaller* than its projection, the formula has been used upside down.

## 02 Worked examples

**EXAMPLE 1** A plane passes through the midpoints of six edges of a cube of edge  $a$ . Name the section and find its perimeter.

- ① Six faces crossed, so a hexagon.
- ② Each side joins two midpoints of one face:  $\frac{a\sqrt{2}}{2}$ .
- ③ All six equal — a regular hexagon.
- ④ Perimeter  $3a\sqrt{2}$

ANSWER

A regular hexagon of perimeter  $3a\sqrt{2}$

**EXAMPLE 2** Find the area of the diagonal section of a cube of edge 6.

- ① It is the rectangle through two opposite edges.
- ② Sides 6 and  $6\sqrt{2}$ .
- ③  $S = 36\sqrt{2} \approx 50.9$

ANSWER

$36\sqrt{2} \approx 50.9$

**EXAMPLE 3** A section of area  $S$  makes  $60^\circ$  with the base. Find its projection.

$$① \quad S_{proj} = S \cos 60^\circ$$

$$② \quad = 0.5 S$$

③ Half the section.

ANSWER

$$0.5 S$$

**EXAMPLE 4** A plane through  $A$ ,  $B$  and  $C_1$  cuts a cube of edge 6. Find the area of the section.

① Projection on the base: triangle  $ABC$ , area 18.

② Dihedral angle along  $AB$  is  $45^\circ$ .

$$③ \quad S = \frac{18}{\cos 45^\circ}$$

$$④ \quad = 18\sqrt{2} \approx 25.5$$

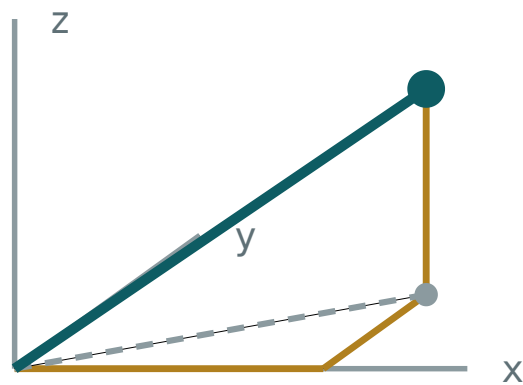
ANSWER

$$18\sqrt{2} \approx 25.5$$

### 03 Interactive model

Cut a potato or a block of foam along a plane through three marked points; the class predicts the polygon before the cut.

## Cutting the box



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

### 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH                 | O'ZBEKCHA            | РУССКИЙ                |
|-------------------------|----------------------|------------------------|
| Section (cross-section) | Kesim                | Сечение                |
| Cutting plane           | Kesuvchi tekislik    | Секущая плоскость      |
| Trace                   | Iz                   | След                   |
| Polyhedron              | Ko'pyoq              | Многогранник           |
| Face                    | Yoq                  | Грань                  |
| Edge                    | Qirra                | Ребро                  |
| Orthogonal projection   | Ortogonal proyeksiya | Ортогональная проекция |
| Projection of an area   | Yuza proyeksiyasi    | Проекция площади       |
| Diagonal section        | Diagonal kesim       | Диагональное сечение   |
| Midpoint section        | O'rta kesim          | Среднее сечение        |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Each side of a section lies in:

two faces

exactly one face

the base

no face

A cutting plane meets two parallel faces in:

perpendicular lines

parallel lines

one point

nothing

A cube's section has at most:

4 sides

5 sides

6 sides

8 sides

A tetrahedron's section has at most:





$S_{\text{projection}}$  equals:





A section through six edge-midpoints of a cube is:





## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A plane parallel to a face of a cube cuts:

ANSWER A square

2. A corner cut off a cube gives:

ANSWER A triangle

3. Maximum number of sides of a cube section

ANSWER 6

4. Maximum number of sides of a tetrahedron section

ANSWER 4

5. Side of the equilateral section through three face-diagonals of a cube of edge  $a$

ANSWER  $a\sqrt{2}$

6. Diagonal section of a cube of edge 6

ANSWER  $36\sqrt{2}$

7.  $S_{\text{projection}}$  when  $S = 20$ ,  $\theta = 60^\circ$

ANSWER 10

Medium · the standard question

7 PROBLEMS

1. Perimeter of the regular hexagonal section of a cube of edge  $a$

ANSWER  $3a\sqrt{2}$

2. Area of that hexagon for  $a = 4$

ANSWER  $12\sqrt{3} \approx 20.8$

3. Area of the equilateral section through three face-diagonals,  $a = 6$

ANSWER  $18\sqrt{3} \approx 31.2$

4. Section through  $A, B, C_1$  of a cube of edge 6

ANSWER  $18\sqrt{2} \approx 25.5$

5.  $S_{\text{section}}$  when  $S_{\text{proj}} = 12, \theta = 45^\circ$

ANSWER  $12\sqrt{2} \approx 17.0$

6. How many faces does a pentagonal section cross?

ANSWER 5

7. A plane parallel to two opposite edges of a cube cuts:

ANSWER A rectangle

### Hard · stretch and reason

7 PROBLEMS

1. Prove that the hexagonal section through the six edge-midpoints of a cube is regular

ANSWER Each side is  $\frac{a\sqrt{2}}{2}$ , angles  $120^\circ$

2. Area of the section of a cube of edge  $a$  through one edge and the midpoint of the opposite edge

ANSWER  $\frac{a^2\sqrt{5}}{2}$

3. A plane cuts a cube of edge 4 through three vertices of one face and one of the opposite: name the section

ANSWER A quadrilateral (a trapezium)

4. A section of a cube has area  $12\sqrt{2}$  and projects to 12: find  $\theta$

ANSWER  $45^\circ$

5. A regular tetrahedron of edge  $a$ : the area of the section through one edge and the midpoint of the opposite

ANSWER  $\frac{a^2\sqrt{2}}{4}$

6. A plane parallel to the base cuts a pyramid halfway up: the ratio of the areas

ANSWER  $1 : 4$

7. Show that the section of a cube through three face-diagonals meeting at a vertex is equilateral

ANSWER All three sides are  $a\sqrt{2}$

## 07 Homework

### Homework — 5 tasks

Draw each cube large, and mark the three given points before drawing any line.

1. State the three rules for constructing a section, with a sketch for each.

2. Construct the section of a cube by the plane through the midpoints of six edges, and name it.
3. Find the area of the diagonal section of a cube of edge 8 cm.
4. A section makes  $30^\circ$  with the base and its projection has area  $9 \text{ cm}^2$ . Find the area of the section.
5. Explain in two sentences why the section of a tetrahedron can never be a pentagon.